TASK 5:

Ticket class:

app > ticket.py > Ticket

```python
from dataclasses import dataclass
from datetime import date
import sqlite3

@dataclass
class Ticket:
    ticket_id: str
    student_number: str
    student_name: str
    category: str
    description: str
    priority: str
    status: str = "Open"
    date_logged: str = str(date.today())
    closing_comment: str = ""

def generate_ticket_id():
    conn = sqlite3.connect("data/tickets.db")
    cursor = conn.cursor()
    cursor.execute("SELECT ticket_id FROM tickets ORDER BY ticket_id DESC LIMIT 1")
    last = cursor.fetchone()
    conn.close()

    if last:
        num = int(last[0][1:]) + 1
        return f"T{num:03d}"
    else:
        return "T001"

```

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Sample database:

database.py M × | tickets.db U | ticket.py M | main.py M | __init__.py U

app > database.py > init_db

You, 41 minutes ago | 1 author (You)

```python
import sqlite3

def init_db():
    conn = sqlite3.connect("data/tickets.db")
    cursor = conn.cursor()

    cursor.execute("""
                        CREATE TABLE IF NOT EXISTS tickets (
                        ticket_id TEXT PRIMARY KEY,
                        student_number TEXT NOT NULL,
                        student_name TEXT NOT NULL,
                        category TEXT NOT NULL,
                            description TEXT NOT NULL,
                        priority TEXT NOT NULL,
                            status TEXT NOT NULL,
                        date_logged TEXT NOT NULL,
                        closing_comment TEXT
                        );
                        """)

    conn.commit()          You, 41 minutes ago • Uncommitted changes
    conn.close()

if __name__ == "__main__":
    init_db()
    print("Database initialized successfully.")
```

This structure is suitable for the tickets and all necessary data types are chosen for the table for example student ID must start with ST so I made it text and the same with ticket ID.